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Anal Roy Chowdhury

Ph.D. Student · IACS Kolkata · Machine Learning & Sign Language Processing

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Education

Ph.D., Mathematical & Computational Sciences, IACS Kolkata — Aug 2023 – Present
Supervisor: Dr. Debarshi Kumar Sanyal

M.Sc., Electronic Science, Acharya Prafulla Chandra College (WBSU) — 2016 – 2018
Thesis: Electrical Properties of ID-DG & TG-DG MOSFETs

B.Sc., Physics (Hons.), Seth Anandram Jaipuria College, University of Calcutta — 2013 – 2016

Research Experience

Ph.D. Scholar — SMCS, IACS Kolkata — Aug 2023 – Present

- Developing multimodal & motion-aware deep learning systems for Indian Sign Language (ISL) recognition and translation.

Project Linked Personnel — ISI Kolkata (under Dr. Swagatam Das) — Jul 2021 – Mar 2023

- Led research on multi-view clustering using multiple kernel methods; published a comprehensive survey in *Information Fusion* (2025, Elsevier).

Research Project Assistant — Sangutech Enterprises Pvt. Ltd. (DBT-BIRAC funded) — Feb – Dec 2019

- Built deep learning pipelines for biomedical image analysis; co-invented UV-light platelet aggregation detection system → **Indian Patent IN580844 B1**.

Selected Publications & Patent

1. **Anal Roy Chowdhury**, A. Gupta, S. Das. "Deep multi-view clustering: A comprehensive survey." *Information Fusion*, 2025. DOI: 10.1016/j.inffus.2025.103012
2. **Anal Roy Chowdhury** and D.K. Sanyal. "Can Small Vision–Language Models Perform Sign Language Translation?" *Findings of ACL 2026*, San Diego.
3. **Anal Roy Chowdhury** and D.K. Sanyal. "Enhancing Indian Sign Language Translation via Motion-Aware Modeling." *WSLP @ AACL-IJCNLP 2025*, IIT Bombay.
4. [**Patent**] S. Deb, . . . , **A. Roy Chowdhury** et al. *Agonist-free system for detecting platelet function.* **Indian Patent IN580844 B1**, granted 17 Feb 2026.

Technical Skills

Languages	Python, MATLAB, C
Frameworks	PyTorch, Scikit-learn, OpenCV
Areas	Deep Learning, Multi-view Clustering, Vision–Language Models, Sign Language NLP
Tools	Git/GitHub, LaTeX, Jupyter Notebook

Awards, Achievements & Service

- **UGC-NET (LS)** qualified — Electronic Science (Dec 2020 & Jun 2021 merged cycle).
- **Student Travel Grant** — CODS-COMAD 2024 (IIT Jodhpur) & CODS-COMAD 2023 (IIIT Bangalore).
- **Administrative Chair**, Winter School on Deep Learning 2023, ISI Kolkata; **Organizing Member**, WSDL 2022.
- **Session Instructor** — Hands-on CNN (PyTorch) & ML (Python) workshops, ISI Kolkata (2022–2023).